CSAI 490 · SENIOR PROJECT I · MILESTONE I

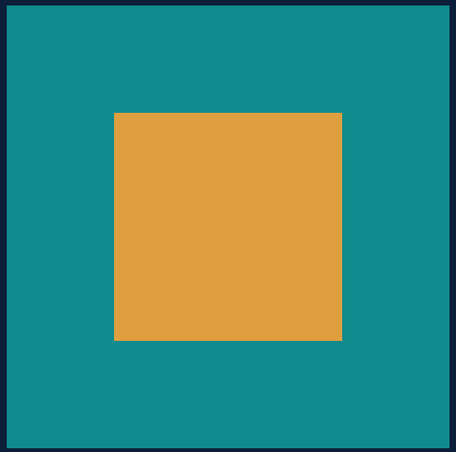
Explainable AI for Cancer Subtype Prediction

Gene expression → molecular subtype prediction → SHAP explanation

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New Giza University · Defense I · Tuesday 27 October 2026



ROADMAP

What we will cover

01

Problem, aim & scope

02

Related work & positioning

03

Dataset & initial EDA

04

Requirements

05

Methodology & technology

06

Risks, plan & dependencies

07

Roles, repository & next steps

CONTEXT

The problem

Breast cancer is heterogeneous.

PAM50 subtypes (Luminal A/B, HER2-enriched, Basal-like, Normal-like) differ in gene expression.

ML can predict them, but black-box predictions are hard to trust without explanation.

GENE EXPRESSION

Different molecular patterns → different subtype signals

Research use only — scientific context, not a clinical diagnosis claim.

PURPOSE

Aim and objectives

Build and evaluate a web-based ML application that predicts breast cancer molecular subtypes from processed RNA-seq gene-expression profiles—and explains predictions in a scientifically qualified way.

01 · Predict

Classify PAM50 subtypes from GSE96058 RNA-seq with LR, SVM, Random Forest and XGBoost.

02 · Evaluate

Compare models and feature sets (PAM50, ANOVA, L1/elastic-net) by macro-F1: 80/20 hold-out + 5-fold CV.

03 · Explain

Use SHAP to show which genes drive each prediction, with scientific qualification.

BOUNDARIES

Scope: in and out

IN SCOPE

Breast cancer PAM50 subtypes (GSE96058)

LR, SVM, Random Forest, XGBoost

Feature selection and macro-F1 evaluation

SHAP explanations

Streamlit app: single/batch upload, PDF report

OUT OF SCOPE

Other cancer types

A clinical diagnostic test

Raw sequencing preprocessing

Treatment recommendation

Causal interpretation of SHAP

LITERATURE REVIEW · 16 REFERENCES

Related work

STUDY	CONTRIBUTION	IMPLICATION FOR THIS PROJECT
Parker et al. – PAM50 [3]	50-gene centroid subtype predictor; reference labelling procedure	PAM50 labels are the target and a benchmark gene panel – not a new subtype.
Cascianelli [6]; Okimoto [7]	Regularized logistic regression is competitive on RNA-seq; reduced PAM50 panels	LR baseline; compare PAM50 vs data-driven gene sets.
Park & Rhee [8]; Shibahara [9]	SHAP-based gene ranking; interpretable patient-level models	Explain with SHAP; present as model explanation, not causality.
Veerla [15]; Raj-Kumar [11]	PAM50 assignment stability; PCAPAM50 vs IHC groups	Document label provenance; assess Normal-like separately.

Contribution: a carefully evaluated, usable system – not a new subtype, a clinical test, or novel XAI. Full 16-reference review available.

POSITIONING

Scientific basis in. Usable integration out.

Existing work provides

the scientific basis

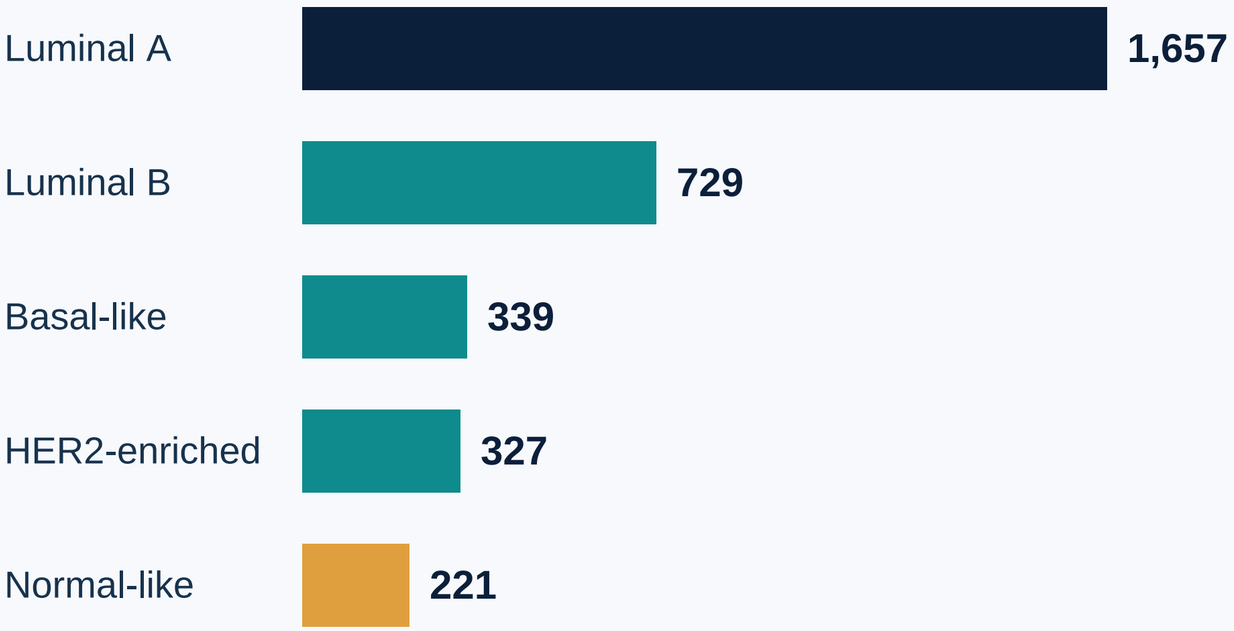
Our contribution is

carefully evaluated, usable integration

Not a new subtype. Not a clinical test.

DATA

Dataset: GSE96058 (SCAN-B)



Samples per PAM50 subtype (n = 3,273) – Luminal A is about half

Population-based breast cancer RNA-seq

NCBI GEO · GSE96058 (SCAN-B)

3,273 primary tumors
(136 technical replicates excluded)

30,865 genes · log2(FPKM + 0.1)

50/50 PAM50 genes represented

49 exact + 1 alias (ORC6L = ORC6)

EXPLORATION — NOTEBOOK OBSERVATIONS, NOT MODEL RESULTS

Initial EDA: what we found and verify next

Samples and genes

3,409 columns → 3,273 primary samples after excluding 136 replicates; 30,865 genes, log2(FPKM + 0.1).

NOTEBOOK

Label distribution

Imbalanced: Luminal A 1,657 (~51%), Luminal B 729, Basal-like 339, HER2 327, Normal-like 221.

NOTEBOOK

PAM50 gene coverage

All 50 PAM50 genes represented after ORC6L to ORC6 alias resolution; no missing or duplicate values.

VERIFIED

Leakage safeguards

Map replicates to patients; split by patient; fit selection and scaling in training folds only.

VERIFY NEXT

SYSTEM BEHAVIOR

Functional requirements

FR-01 · 02 · 03

Input & validation

01 Single / batch input

02 Input validation

03 Saved preprocessing

FR-04 · 05 · 11

Prediction

04 Subtype prediction

05 Class scores

11 Clear error feedback

FR-06 · 07 · 08

Explanation

06 Sample SHAP plots

07 Global SHAP summary

08 Gene evidence + links

FR-09 · 10 · 12

Interface & reporting

09 Streamlit results

10 PDF report

12 Research-use notice

QUALITY ATTRIBUTES

Non-functional requirements

NFR-01 – 03

Rigor & reproducibility

Macro-F1 on held-out data; patient-level split; seeds and versions logged

NFR-04 – 06

Integrity & validity

Versioned schema; SHAP matches deployed model; traceable sources

NFR-07 – 09

Usability & performance

No-code workflow; 30 s per sample (target); graceful error handling

NFR-10 – 12

Security & maintainability

Uploads not retained; modular code; tested in mainstream browsers

SEPARATION BY DESIGN

Methodology pipeline

OFFLINE DEVELOPMENT

GSE96058 + PAM50 labels — Patient-level split — Feature sets + 5-fold CV — Held-out evaluation — Export versioned model

USER-FACING APPLICATION

Upload profile(s) — Validate schema — Saved pipeline → predict — SHAP + gene evidence — Results + PDF report

The classifier is never retrained when a user uploads a sample.

IMPLEMENTATION CHOICES

Methods and technology stack

Scikit-learn · XGBoost

Pipelines, feature selection, CV; LR, SVM, Random Forest, XGBoost.

SHAP

Sample-level and model-level gene contributions.

Streamlit

Single/batch upload, results and PDF report.

Python · Pandas · NumPy

Data preparation and application logic.

Git + GitHub

Source control, team collaboration, version history.

Jupyter · Matplotlib · Seaborn

EDA, experiments and evaluation figures.

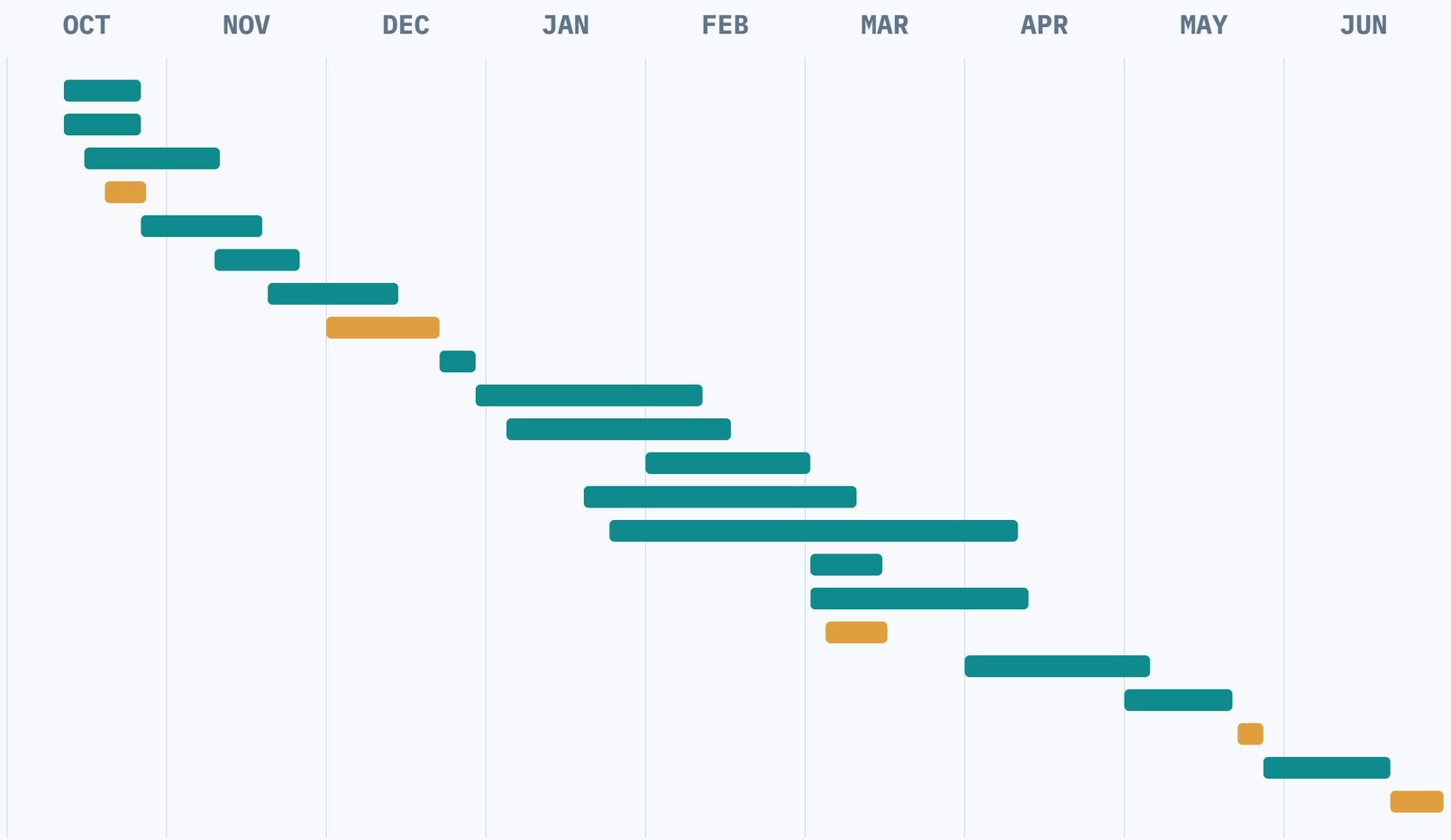
DELIVERY CONTROL

Risk log

ID	RISK	LIKELIHOOD	IMPACT	MITIGATION
R1	Data leakage inflates scores	Medium	High	Split by patient first, keep replicates together, fit selection inside training folds, test once.
R2	Weak subtypes, e.g. Normal-like	High	High	Per-class metrics, confusion analysis; justify any class change scientifically.
R3	Incompatible uploaded data	High	High	Versioned feature schema; reject bad files with clear messages.
R4	Misleading SHAP or gene claims	Medium	High	Check gene-ranking stability; keep model attributions separate from verified evidence.
R5	Schedule delays	Low	High	Track milestones in GitHub; get an end-to-end baseline working early.

OCTOBER 2026 → JUNE 2027

Project plan and Gantt chart



T01	Scope, requirements, risk plan
T02	Literature review and references
T03	Dataset, labels, replicate audit
T04	Defense I materials and repository
T05	Scale, PAM50 alias and EDA checks
T06	Reproducible split and eval protocol
T07	Initial PAM50-gene baselines
T08	Initial thesis and Defense II work
T09	Advisor feedback, revised thesis
T10	PAM50 vs ANOVA gene comparison
T11	L1 / elastic-net gene panels
T12	LR, SVM, RF, XGBoost comparison
T13	First Streamlit UI and validation
T14	Curated gene evidence and refs
T15	Final model and held-out test
T16	SHAP explanations, reliability
T17	Defense III model and app demo
T18	Integration and PDF reporting
T19	Security, usability, speed tests
T20	Pre-final demo and refinements
T21	Final thesis, GitHub audit
T22	Final defense preparation

Planned work, not completed results. Amber = defense-linked tasks. Milestones: Defense I 27 Oct · Initial thesis 15 Dec · Defense II 22 Dec · Revised thesis 29 Dec · Defense III 16 Mar · Pre-final 23–27 May · Final thesis 20 Jun · Final defense 27–30 Jun

CRITICAL PATH

Key dependencies and milestones



Milestones

Defense I: 27 Oct 2026

Initial thesis: 15 Dec · Defense II: 22 Dec · Revised thesis: 29 Dec

Defense III: 16 Mar 2027 · Pre-final: 23–27 May

Final thesis: 20 Jun · Final defense: 27–30 Jun

DATES TO BE CONFIRMED WITH THE UNIVERSITY

TEAM ALLOCATION

Roles and responsibilities

Ziad Elmorsy

Scope, requirements, risk plan
EDA checks, split protocol
L1 / elastic-net panels
Streamlit UI, final model
SHAP reliability

Mohamed Samy

Dataset and replicate audit
Split protocol
L1 / elastic-net panels
Streamlit UI
Gene evidence
Security and speed tests

Mostafa Elzeiny

Dataset and replicate audit
EDA and alias checks
PAM50 baselines
ANOVA comparison
Final model evaluation
Testing

Gamal Mohamed

Literature and references
PAM50 baselines
ANOVA comparison
Gene evidence
SHAP explanations

ALL FOUR TOGETHER · Defense materials, thesis, model comparison, integration, demos. Pairings show practical allocation, not leadership.

EVIDENCE OF STARTED WORK

Repository and progress so far

DONE / EVIDENCE AVAILABLE

Specification, requirements, risk plan

Literature review (16 references)

GSE96058 EDA: 50/50 PAM50 genes verified

Project plan and Gantt

GitHub repository (next slide)

NEXT

Define final model input feature schema

Group-safe 80/20 split and CV protocol

PAM50-gene LR and SVM baselines

First Streamlit prototype

EDA figures are notebook observations still being verified. No model results yet.

REPOSITORY

Explore the project on GitHub

GitHub Link

Code, notebooks, documentation and project plan.

IMMEDIATE NEXT STEPS

From inspected data to evaluated application

1 Verify notebook observations and class counts
2 Freeze the feature and label contract
3 Finalize the evaluation protocol
4 Build the offline pipeline and save artifacts
5 Integrate prediction + SHAP in Streamlit

Thank you.

Questions?

Results remain planned / not yet achieved unless explicitly shown otherwise.